Where to park K clinic vans: clustering and the k-means algorithm

BIS 527: Introduction to Health Data Science Lecture 10, Tuesday, October 6, 2026

In one paragraph

A health department with K mobile clinic vans must decide where to park them for the households of a city, and each household will go to its nearest van. Choosing the parking spots splits the households into K groups, so it is a clustering question. We state the clustering question in general, narrow it to the K-centers question, in which each of K centers (the vans) serves one cluster, and narrow that to the k-means problem, which scores a choice by the within-cluster sum of squares W. The k-means algorithm alternates two parts: it partitions the households by their nearest centers, and it learns new centers as the means of the clusters. It always stops, but the split where it stops can be arbitrarily far from the best one, which we prove. We end with two surveys of the same city, Snow's cholera map, and a recent attempt to find groups of sepsis patients again in other hospitals. In each, a correct algorithm returns an answer that depends on which data it receives, and judging such answers is the work of the next part of the course.

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Notation

Every symbol and term used in these notes

Units and points	
n, i	The number of units (here, points), and a unit. Units are numbered $i = 0, 1, \ldots, n-1$.
p, j	The number of measurements per unit, and a measurement. Measurements are numbered $j = 0, \ldots, p-1$.
x_i, $x_i[j]$	Point i, the measurements of unit i: $x_i = (x_i[0], \ldots, x_i[p-1])$, and its coordinate j. In the household example, $p = 2$, and $x_i[0]$ and $x_i[1]$ are how far east and how far north household i is, in units of 100 m.
$\|x - y\|^2$	The squared distance between points x and y: $\sum_{j=0}^{p-1}(x[j] - y[j])^2$.
$\|x - y\|$	The straight-line distance, $\sqrt{\|x - y\|^2}$.
$d(x, y)$	A dissimilarity: a number $d(x, y) \geq 0$ for each pair of points, with $d(x, x) = 0$ and $d(x, y) = d(y, x)$. Larger means less similar.

Clusterings and splits	
C	A clustering: nonempty sets of points $C_0, \ldots, C_{K-1}$, no two overlapping (Definition 1).
$U(C)$	The unclustered points: those in none of the sets of C.
within, between pair	Two points in the same cluster, and two points in different clusters.
well separated	Every within pair has a smaller dissimilarity than every between pair (Definition 1(c)).
cohesion, separation	Small dissimilarity within clusters, and large dissimilarity between them.
Q	A clustering criterion: a number for each clustering. The clustering question maximizes it (Definition 1(d)).
A1 to A4	The four assumptions that narrow Definition 1 to Definition 2 (Section 3).
K, k	The number of clusters, and a label. Labels are $k = 0, \ldots, K-1$.
$\mathbf{1}[E]$	The indicator of a statement E: 1 if E is true, 0 if not.
z_i, z	The label of point i: $z_i := k$ when $\mathbf{1}[x_i \in C_k] = 1$. The list $z = (z_0, \ldots, z_{n-1})$ is an assignment.
C_k	Cluster k, a set of points. We write $x_i \in C_k$ or $i \in C_k$, and $C_k = \{i : z_i = k\}$. It may be empty.
$\|S\|$	The number of elements of a set S.
$\bar{x}_S$	The mean of a nonempty set S of points: the point with coordinate j equal to $\frac{1}{\|S\|}\sum_{i \in S} x_i[j]$.
$\bar{x}$	The mean of all n points (the remark after Claim 1).
μ, μ_k	Centers: a list $\mu = (\mu_0, \ldots, \mu_{K-1})$ of K points, one per label. After a learning step, μ_k is the centroid of C_k, its mean $\bar{x}_{C_k}$.
$D(\mu)$	The total cost of the K-centers question: $\sum_i \min_k d(x_i, \mu_k)$, each point paying for its nearest center (Definition 2).
$W(\mu)$	The k-means cost of centers μ: $\sum_i \min_k \|x_i - \mu_k\|^2$, each point at its nearest center (Definition 3).
$W(z, \mu)$	The within-cluster sum of squares of assignment z with centers μ: $\sum_i \|x_i - \mu_{z_i}\|^2$ (Definition 3). It is at least $W(\mu)$.
split	An assignment with the labels' names ignored. $\{0, 1, 2\}$ and $\{3, 4, 5\}$ is one split.
DBSCAN, noise	A clustering method based on density, and a point it puts in no cluster (Sections 3 and 11).

The algorithm

k-means	The algorithm of Section 4. It alternates a partition step and a learning step (Section 4).
round	One partition step, followed by one learning step unless the algorithm stops.
tie rule	In the partition step, a point keeps the current label when that label's center is among the nearest.
starting centers	The centers before the first round. In the example, the points of households 0 and 1.
$z^{(r)}$, $\mu^{(r)}$	The labels after partition step r, and the centers after learning step r. $\mu^{(0)}$ are the starting centers. Rounds are numbered from 1.
W_r	$W(z^{(r)}, \mu^{(r)})$, the within-cluster sum of squares after learning step r (proof of Claim 3).
W^*	The value of the k-means problem: the smallest $W(\mu)$ over all centers (Section 3).
approximation gap	The ratio of the W that k-means returns to W^*. Claim 4 shows it has no upper bound.
G, L	In Claim 4, any positive number, and the width of the rectangle whose gap is $L^2 > G$.

Populations and samples (Section 7)

population	The set of all points the question is about. Here, a city of $N = 24$ points.
sample	The points whose data we have. Here, the six points of one survey.
best split	A split with the smallest W, with the cluster means as centers.
population centers	The centers of the best split of the whole population.
sample centers	The centers of the best split of a sample.
reported centers	The centers that k-means returns for a sample.

Snow's map (Section 8)

Voronoi diagram	The regions of points nearest each of a set of fixed centers.
walking distance	The length of a shortest route along the streets.
even spread	Points spread uniformly over a region, with no place favored.
denominator	The number of people at risk. A count of deaths divided by it gives a rate.

Cost (Section 11)

Θ	As in Lecture 4: a cost is $\Theta(f)$ when, for all large sizes, it lies between two positive constant multiples of f.

1 Where to park K clinic vans

A clustering question, and six households

In this section, we pose the question of the lecture with a hypothetical planning problem.

- A health department has K mobile clinic vans, for vaccination or screening, and must decide where to park them. The units are households spread across a city or a region.
- Each household will go to its nearest van. So once the vans are parked, the households fall into K groups, one per van, and a good choice keeps every household close to its own van.
- Choosing where to park the vans is therefore a clustering question: split the households into K groups and place one van in each, so that households are close to their van. A van's parking spot is the center of its cluster.
- No household comes with a group label. The groups have to be found from where the households are. Other questions have the same shape, such as community detection in the contact networks of Lectures 5 and 6, which looks for groups of people with many contacts within and few between (Salathé and Jones, 2010).
- To work by hand, we use six households of a survey A, each a point on a map in units of 100 m (Figure 1). The data are synthetic.

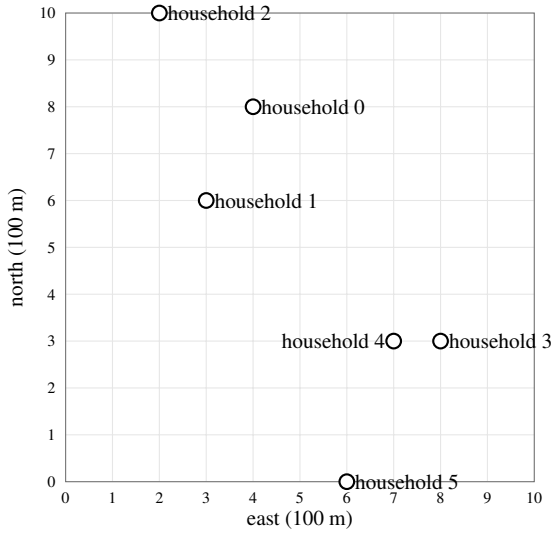


Figure 1. Survey A: the homes of six synthetic households (east, north), in units of 100 m.

- By eye, households 0, 1 and 2 form one neighborhood and households 3, 4 and 5 another, so with $K = 2$ one van would park in each.
- Two questions follow. What is a clustering, stated precisely? And how do we compute a good one for thousands of households, where no figure can be drawn? Section 3 answers the first question, and Sections 4 to 6 answer the second.

2 Setup

Points, distances, labels and means

In this section, we fix the vocabulary that the definitions of Section 3 are written in.

Prerequisites. We use sums, sets and functions, and the counting arguments of Lectures 2 to 8.

Points. The data are n units, numbered $i = 0, 1, \ldots, n-1$, each with p measurements. We collect the measurements of unit i in a *point* $x_i = (x_i[0], x_i[1], \ldots, x_i[p-1])$, with positions counted from 0, as in Python, and from here on we call unit i simply point i. In the example of survey A, the units are households, $n = 6$ and $p = 2$: the coordinate $x_i[0]$ is how far east household i is and $x_i[1]$ is how far north, in units of 100 m, so $x_0 = (4, 8)$ and $x_5 = (6, 0)$.

Squared distance. For two points x and y with p coordinates each, the *squared distance* is

$$\|x - y\|^2 := \sum_{j=0}^{p-1} \big(x[j] - y[j]\big)^2.$$

It is the square of the straight-line distance between the two points. For example, $\|x_0 - x_5\|^2 = (4-6)^2 + (8-0)^2 = 4 + 64 = 68$. The straight-line distance itself is $\|x - y\| := \sqrt{\|x - y\|^2}$, so $\|x_0 - x_5\| = \sqrt{68}$, a little over 8.

Dissimilarity. A *dissimilarity* is a rule d that gives every pair of points x and y a number $d(x, y) \geq 0$, with $d(x, x) = 0$ and $d(x, y) = d(y, x)$. A larger value means less similar. The straight-line distance and the squared distance are both dissimilarities, and so is the walking distance along streets in Section 8. On a contact network, the number of contacts on a shortest path between two people, which BFS computes (Lecture 6), is a dissimilarity between people too. Choosing d is part of stating the question, and Section 3 shows which choice k-means makes.

Clusters. Fix a number of clusters K. A split of the points into K clusters is a list of sets $C_0, \ldots, C_{K-1}$ of points that do not overlap and together contain every point. We write $x_i \in C_k$, or simply $i \in C_k$, for "point i is in cluster k". A cluster may be empty. We write $|S|$ for the number of elements of a set S.

Indicator. For a statement E, the *indicator* $\mathbf{1}[E]$ is 1 if E is true and 0 if it is false. For example, $\mathbf{1}[x_i \in C_k]$ is 1 exactly when point i is in cluster k.

Labels. The *label* of point i is the number of its cluster,

$$z_i := k \quad \text{when} \quad \mathbf{1}[x_i \in C_k] = 1.$$

Every point is in exactly one cluster, so every point has exactly one label, a number in $\{0, \ldots, K-1\}$. The list $z = (z_0, \ldots, z_{n-1})$ is an *assignment*: in the language of functions, it maps the points $\{0, \ldots, n-1\}$ to the labels $\{0, \ldots, K-1\}$. The labels and the clusters carry the same information, because

$$C_k = \{i : z_i = k\},$$

read "the set of all i such that $z_i = k$". For survey A with $K = 2$, the upper-left and lower-right split has $C_0 = \{0, 1, 2\}$ and $C_1 = \{3, 4, 5\}$, so $z = (0, 0, 0, 1, 1, 1)$. The assignment $(0, 0, 0, 0, 0, 0)$ has $C_1 = \{\}$.

Means. For a nonempty set S of points, the *mean* $\bar{x}_S$ is the point whose coordinate j is the average of the coordinates j of the points in S:

$$\bar{x}_S[j] := \frac{1}{|S|} \sum_{i \in S} x_i[j] \qquad \text{for } j = 0, \ldots, p-1.$$

For $S = \{0, 2\}$ in survey A, $\bar{x}_S = \left(\frac{4+2}{2}, \frac{8+10}{2}\right) = (3, 9)$.

3 From the clustering question to the k-means problem

Three definitions, each narrower than the last

In this section, we state the clustering question in general, and then narrow it in two steps until it becomes a problem with a single number to minimize. Each step adds assumptions, and each assumption sets some clustering methods aside.

Definition 1 (clustering, and the clustering question)

Let $x_0, \ldots, x_{n-1}$ be n points, numbered $0, \ldots, n-1$, and let d be a dissimilarity.

(a) A *clustering* is a collection $C = \{C_0, \ldots, C_{K-1}\}$ of sets of points, for some $K \geq 1$, such that $C_k \neq \{\}$ for every k, and $C_k \cap C_l = \{\}$ for every $k \neq l$. The points in $U(C) := \{0, \ldots, n-1\} \setminus (C_0 \cup \cdots \cup C_{K-1})$ are *unclustered*.

(b) A *within pair* of C is a set $\{i, i'\}$ of two points in the same cluster. A *between pair* is a set $\{j, j'\}$ of two points in different clusters.

(c) C is *well separated* if $d(x_i, x_{i'}) < d(x_j, x_{j'})$ for every within pair $\{i, i'\}$ and every between pair $\{j, j'\}$.

(d) A *clustering criterion* is a function Q that gives every clustering C a number $Q(C)$, computed from the points and d. For a chosen criterion Q, the *clustering question* asks for a clustering C with the largest $Q(C)$.

Part (a) says what a clustering is: groups that are not empty and do not overlap, with some points possibly left out. Part (c) states formally what we would like a clustering to achieve, *cohesion* and *separation* at once: any two points in one cluster are more similar than any two points in different clusters. Part (c) cannot be the whole question, for two reasons. Every set of points has two well-separated clusterings that say nothing, one cluster holding everyone ($K = 1$, so there are no between pairs) and n clusters of one point each (so there are no within pairs). And real data often have no well-separated clustering in between. A method therefore has to choose a criterion Q that trades cohesion against separation, and part (d) leaves that choice open. Methods differ in their criterion. Some, such as DBSCAN, define their clustering directly by a rule instead (Section 11): DBSCAN puts two points in the same cluster when a chain of densely packed points links them, so K is an output, a cluster can have any shape, and points in sparse regions stay unclustered. Hierarchical clustering returns a whole tree of nested clusterings, one for every K. Mixture models give each point a probability of belonging to each group, so their output becomes a clustering in the sense of part (a) only once each point is given its most probable group.

Five candidate outputs for survey A. The table checks five outputs against parts (a) and (c), with d the squared distance.

	Sets	A clustering?	Well separated?
(i)	$\{0, 1, 2\}$ and $\{3, 4, 5\}$	yes	yes: the largest within value, 17 for $\{1, 2\}$, is below the smallest between value, 25 for $\{1, 4\}$
(ii)	$\{0, 1, 2\}$ and $\{3, 4\}$	yes, household 5 unclustered	yes: 17 is below 25 again
(iii)	$\{0, 1, 2, 3\}$ and $\{3, 4, 5\}$	no, household 3 is in two sets	does not apply
(iv)	$\{0, 1, 2, 3, 4, 5\}$	yes, $K = 1$	yes, because there are no between pairs
(v)	$\{0, 2\}$ and $\{1, 3, 4, 5\}$	yes	no: within pair $\{1, 5\}$ has 45, between pair $\{0, 1\}$ has 5

The K-centers question

Choose K centers, and send each point to the nearest

We now narrow Definition 1 with four assumptions.

A1 (Fixed number of clusters.) K is chosen in advance.

A2 (Centers.) There are K *centers* $\mu_0, \ldots, \mu_{K-1}$, points with p coordinates that need not be any of the data points.

A3 (Nearest center.) Every point belongs to the cluster of its nearest center, so the centers determine the clusters.

A4 (Cost.) Point i pays the dissimilarity to its nearest center, and the centers are judged by the total payment.

Definition 2 (the K-centers question)

Input: points $x_0, \ldots, x_{n-1}$, a dissimilarity d, and a number of clusters K.
Output: centers $\mu = (\mu_0, \ldots, \mu_{K-1})$ with the smallest total cost

$$D(\mu) := \sum_{i=0}^{n-1} \min_{k=0,\ldots,K-1} d\big(x_i, \mu_k\big).$$

In words: we choose K points, each point pays the dissimilarity to the nearest of them, and we want the choice with the smallest total. The clusters then come with the centers: cluster k holds the points whose nearest center is μ_k. For the vans of Section 1, K is the number of vans (A1), μ_k is where van k parks (A2), each household goes to its nearest van (A3), and pays the dissimilarity between its home and that van (A4). Assumptions A1 and A3 set DBSCAN aside, because it fixes no K and leaves some points unclustered, and in the table above A3 rules out output (ii). The cost D measures cohesion only. Separation, aim (b) of Definition 1, does not appear in it, and the remark after Claim 1 shows that for k-means it is obtained anyway.

Definition 2 is still a family of problems, one for each dissimilarity, and the vans show why the choice matters. If the department cares about the total distance travelled, it uses the straight-line distance, and Definition 2 is the *k-median* problem. If each van must park at one of the points, it becomes the *k-medoids* problem. If the department cares only about the longest trip, the total is replaced by the largest single payment, and it becomes what the algorithms literature calls the *k-center* problem.

The k-means problem

Where to put K centers, with squared distances

Definition 3 (the k-means problem, and the within-cluster sum of squares)

(a) The *k-means problem* is Definition 2 with the squared distance as the dissimilarity. For given points $x_0, \ldots, x_{n-1}$ and K, it asks for the centers $\mu = (\mu_0, \ldots, \mu_{K-1})$ with the smallest

$$W(\mu) := \sum_{i=0}^{n-1} \min_{k=0,\ldots,K-1} \|x_i - \mu_k\|^2.$$

The smallest value of $W(\mu)$ is written W^*, the *value* of the k-means problem.

(b) For an assignment z and centers μ, the *within-cluster sum of squares* is

$$W(z, \mu) := \sum_{i=0}^{n-1} \|x_i - \mu_{z_i}\|^2,$$

the total when point i uses center z_i, whether or not it is the nearest one.

In words: the k-means problem asks where to put K centers so that the total squared distance from each point to its nearest center is as small as possible. For the vans, $W(\mu)$ is the total squared distance from households to the vans they use, and the problem asks where to park the vans to make it smallest. Compared with the plain distance, the squared distance weighs points far from their center more heavily: a point twice as far from its center pays four times as much, so for the vans k-means suits a department that wants to avoid very long trips. Claim 1 shows that with this choice the best center of a cluster is its mean, which gives the problem its name.

Part (b) is the quantity the k-means algorithm works with, because the algorithm handles the labels and the centers in separate steps. Each point's term in $W(z, \mu)$ is smallest when z_i is a nearest center, so $W(z, \mu) \geq W(\mu)$ for every assignment, with equality when every point carries the label of a nearest center. So W^* is also the smallest $W(z, \mu)$ over all assignments and all centers.

***W* by cluster.** Adding the same terms cluster by cluster gives the same total,

$$W(z, \mu) = \sum_{k=0}^{K-1} \sum_{i \in C_k} \|x_i - \mu_k\|^2,$$

because every point belongs to exactly one cluster. An empty cluster contributes nothing, whatever its center is.

Three splits of survey A. With $K = 2$ and each center at the mean of its cluster:

	Clusters C_0 and C_1	Centers μ_0 and μ_1	W
(i)	$\{0, 1, 2\}$ and $\{3, 4, 5\}$	$(3, 8)$ and $(7, 2)$	$(1 + 4 + 5) + (2 + 1 + 5) = 18$
(ii)	$\{0, 2\}$ and $\{1, 3, 4, 5\}$	$(3, 9)$ and $(6, 3)$	$(2 + 2) + (18 + 4 + 1 + 9) = 36$
(iii)	$\{0, 1, 2, 3, 4, 5\}$ and $\{\}$	$(5, 5)$ and any point	$10 + 5 + 34 + 13 + 8 + 26 = 96$

In (i), household 0 contributes $\|(4, 8) - (3, 8)\|^2 = 1^2 + 0^2 = 1$, household 1 contributes $0^2 + 2^2 = 4$, and household 2 contributes $1^2 + 2^2 = 5$. Split (i) has the smallest W of the three, which matches the split most readers see in Figure 1.

Splits. Two assignments that differ only in the names of the labels, such as $(0, 0, 0, 1, 1, 1)$ and $(1, 1, 1, 0, 0, 0)$, put the points in the same groups. A *split* is an assignment with the names of the labels ignored, here $\{0, 1, 2\}$ and $\{3, 4, 5\}$. A *best split* is a split with the smallest W when each center is the mean of its cluster.

How many assignments? Each of the n points takes one of K labels, so there are K^n assignments. Claim 1 below shows that, for a fixed assignment, the best centers are the cluster means, so trying every assignment solves the problem. For survey A this means $2^6 = 64$ assignments, which is quick. Dropping the two assignments that leave a cluster empty, and counting an assignment and its relabeled copy once, leaves $(64 - 2)/2 = 31$ splits into two nonempty clusters. For 100 points and $K = 2$ it means 2^{100} assignments, more than 10^{30}. Trying all of them finds W^* but takes exponentially many steps, as the plain recursion for the LCS did in Lecture 8, and no method is known that always finds W^* in a number of steps bounded by a polynomial in n and p (Section 11). So we need a different approach.

4 The k-means algorithm

Partition, then learn the new centers

In this section, we first restate the problem that the k-means algorithm aims at, and then state the algorithm and give it in Python.

The problem. The k-means problem (Definition 3) asks, for given points $x_0, \ldots, x_{n-1}$ and a number of clusters K, for the centers $\mu_0, \ldots, \mu_{K-1}$ with the smallest

$$W(\mu) = \sum_{i=0}^{n-1} \min_{k=0,\ldots,K-1} \|x_i - \mu_k\|^2.$$

Its smallest value is W^*. For the vans, the problem asks where to park the K vans, and W^* is the smallest total squared distance from households to their nearest vans that any parking plan can achieve.

The algorithm, as an approximation. The k-means algorithm does not solve this problem exactly. It is a fast procedure that returns centers μ (and the matching assignment) whose W is at least W^* and can be larger, so its answer is an approximation to the answer of the problem. Section 6 proves what the algorithm guarantees and how far its W can be from W^*. The name k-means is used for both the problem and the algorithm, and the two are different things: the problem defines the best split, and the algorithm is one way to look for it.

The k-means algorithm (also called Lloyd's algorithm, after Lloyd 1982) starts from K *starting centers* and repeats *rounds*. Each round has two parts: it partitions the points using the current centers, and then it learns new centers from that partition.

1. **Partition step.** Each point takes the label of a nearest center: z_i becomes a label k with the smallest $\|x_i - \mu_k\|^2$. *Tie rule:* if the point's current label is among the nearest, the point keeps it. Otherwise the point takes the smallest such k. In the first round no point has a label yet, so every point receives one.
2. **Learning step.** Each center becomes the *centroid* of its cluster, the mean of its points' points: μ_k becomes $\bar{x}_{C_k}$ for every nonempty C_k. The center of an empty cluster stays where it is.

Stopping rule. If a partition step changes no label, the algorithm stops at once, without a learning step, and returns z and μ. Receiving a first label counts as a change, so the first round always has a learning step.

The name comes from the learning step: each of the K centers is a mean. (The k in "k-means" is the number of clusters, which we write K, keeping the lowercase k for a label.) The two parts split the work. For fixed centers, the partition step finds the best label for each point. For a fixed partition, the learning step finds the best center for each cluster, computed from the points of that cluster alone. Section 6 shows that each part gives the best value of the quantity it changes (the labels, or the centers), and that this, together with the finite number of assignments, makes the algorithm stop.

In the example, the starting centers are the points of households 0 and 1, with $K = 2$. Software usually picks the starting centers at random, and Section 6 shows that the choice matters. The code below is the algorithm in Python, for any n, p and K. It has one function for each part: `partition` carries out the partition step, `learn_centers` carries out the learning step, and `k_means` repeats the two until the stopping rule applies.

```
def sq_dist(x, c):                          # squared distance
    return sum((x[j] - c[j])**2 for j in range(len(x)))

def mean(points):                           # mean of a nonempty list
    return [sum(x[j] for x in points) / len(points)
            for j in range(len(points[0]))]

def partition(X, centers, z):               # partition step
    new = []
    for i in range(len(X)):
        d = [sq_dist(X[i], c) for c in centers]
        if z[i] is not None and d[z[i]] == min(d):
            new.append(z[i])                # tie rule: keep the label
        else:
            new.append(d.index(min(d)))     # smallest nearest label
    return new

def learn_centers(X, z, centers):           # learning step
    new = []
    for k in range(len(centers)):
        C_k = [X[i] for i in range(len(X)) if z[i] == k]
        if C_k:
            new.append(mean(C_k))
        else:                               # an empty cluster keeps
            new.append(centers[k])          # its center
    return new

def k_means(X, centers):
    z = [None] * len(X)                     # no labels yet
    while True:
        new = partition(X, centers, z)
        if new == z:                        # stopping rule
            return z, centers
        z = new
        centers = learn_centers(X, z, centers)
```

5 A run on survey A

Three rounds, from the points of households 0 and 1

We run k-means on survey A with $K = 2$ and starting centers $\mu_0 = x_0 = (4, 8)$ and $\mu_1 = x_1 = (3, 6)$. In each table, a row gives the squared distances from one household to the two centers, and the smaller one, in bold, gives the label.

Round 1, partition step. Household 2 is nearer to μ_0, and households 3, 4 and 5 are nearer to μ_1.

patient	x_i	to $\mu_0 = (4, 8)$	to $\mu_1 = (3, 6)$	label
0	(4, 8)	**0**	5	0
1	(3, 6)	5	**0**	1
2	(2, 10)	**8**	17	0
3	(8, 3)	41	**34**	1
4	(7, 3)	34	**25**	1
5	(6, 0)	68	**45**	1

Round 1, learning step. $C_0 = \{0, 2\}$ and $C_1 = \{1, 3, 4, 5\}$, so $\mu_0 = \left(\frac{4+2}{2}, \frac{8+10}{2}\right) = (3, 9)$ and $\mu_1 = \left(\frac{3+8+7+6}{4}, \frac{6+3+3+0}{4}\right) = (6, 3)$.

Round 2, partition step. Household 1 now has squared distance 9 to μ_0 and 18 to μ_1, so its label changes from 1 to 0. No other label changes.

patient	x_i	to $\mu_0 = (3, 9)$	to $\mu_1 = (6, 3)$	label
0	(4, 8)	**2**	29	0
1	(3, 6)	**9**	18	**0** (was 1)
2	(2, 10)	**2**	65	0
3	(8, 3)	61	**4**	1
4	(7, 3)	52	**1**	1
5	(6, 0)	90	**9**	1

Round 2, learning step. $C_0 = \{0, 1, 2\}$ and $C_1 = \{3, 4, 5\}$, so $\mu_0 = (3, 8)$ and $\mu_1 = (7, 2)$.

Round 3, partition step. No label changes, so k-means stops and returns split (i) of Section 3, with centers $(3, 8)$ and $(7, 2)$.

patient	x_i	to $\mu_0 = (3, 8)$	to $\mu_1 = (7, 2)$	label
0	(4, 8)	**1**	45	0
1	(3, 6)	**4**	32	0
2	(2, 10)	**5**	89	0
3	(8, 3)	50	**2**	1
4	(7, 3)	41	**1**	1
5	(6, 0)	73	**5**	1

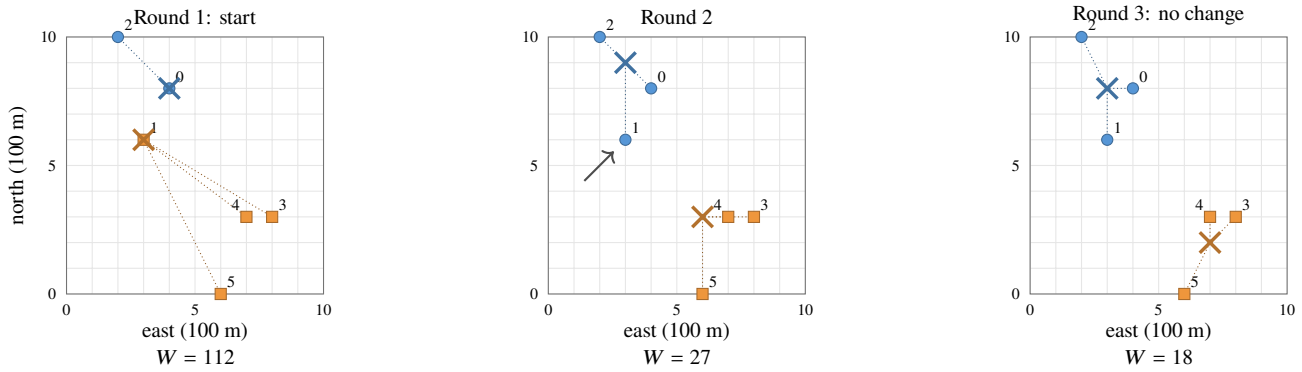


Figure 2. The three partition steps on survey A. Crosses are the centers used in that step.

Circles carry label 0 and squares label 1. A dotted line joins each household to its center, so W is the sum of the squared lengths of the dotted lines. The arrow in round 2 marks household 1, whose label changes. The value of W after each step is

After	partition step 1	learning step 1	partition step 2	learning step 2	partition step 3
W	112	36	27	18	18

Two features of this run are worth noting. First, no step increased W: the partition step lowered it from 36 to 27 by moving one household, and each learning step lowered it by moving the centers. Claim 2 shows that this holds in every run. Second, household 1 began as the starting center of cluster 1 and ended in cluster 0: a starting center is only a first guess, and the household whose point it was can end in the other cluster.

The split where this run stops, with $W = 18$, is the best of all 31 ways to split the six households into two nonempty clusters (checked by computing W for every one of them). Section 6 proves that k-means always stops, and that the split where it stops can be far from the best one.

A larger run

90 households, $K = 3$ vans, eight rounds

Figure 3 runs k-means on 90 synthetic households in three neighborhoods, with $K = 3$ vans and a poor start: the three starting centers are households in the southern neighborhood. In round 1, the partition step gives every northern household to one center, and the learning step moves that center far to the north. In rounds 2 to 6, a second center moves, round by round, toward the north-western neighborhood and takes its households from the first, while the first center slides east. By round 7 each neighborhood has its own center, round 8 changes no label, and k-means stops. W never increases, as Claim 2 requires: it falls from 1449.0 to 163.4, most of it in the first learning step and in rounds 5 to 7. From 300 random choices of three households as starting centers, no run of k-means reached a smaller W than 163.4.

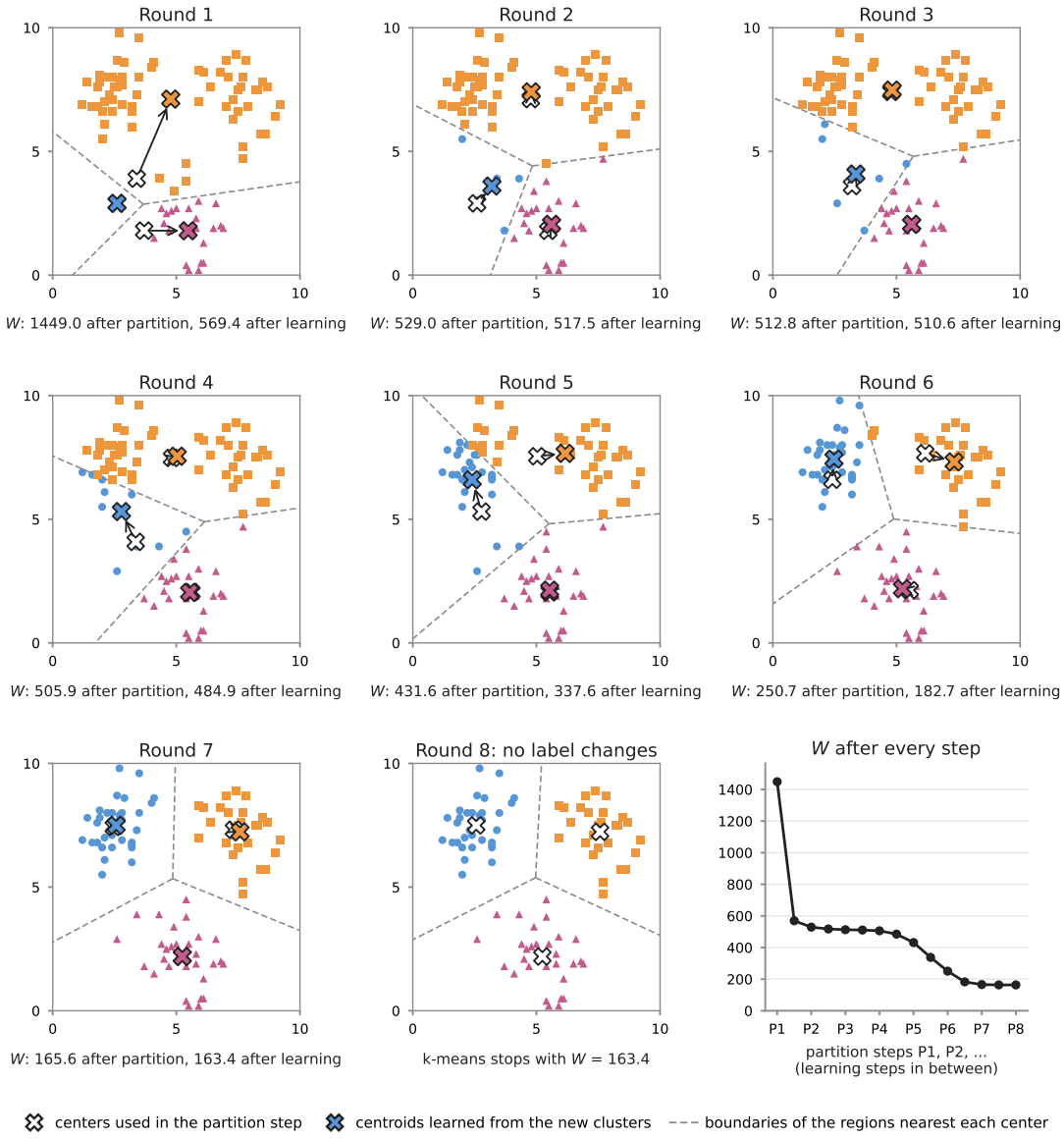


Figure 3. k-means on 90 synthetic households with $K = 3$, one panel per round. Colors and shapes show the labels z_i after each partition step. W is in units of $(100\ \text{m})^2$.

6 What k-means guarantees, and the approximation gap

The best center for one cluster is its mean

In this section, we prove four claims. Claim 1 is about the learning step on its own. Claims 2 and 3 show that W never increases and that the algorithm stops. Claim 4 measures how far the answer of the algorithm can be from W^*, the value of the k-means problem.

Claim 1 (the mean is the best center)

Let S be a nonempty set of points and let μ be any point with p coordinates. Then

$$\sum_{i \in S} \|x_i - \mu\|^2 = \sum_{i \in S} \|x_i - \bar{x}_S\|^2 + |S| \, \|\bar{x}_S - \mu\|^2.$$

So the mean $\bar{x}_S$ gives the smallest sum of squared distances to the points of S, and every other point μ gives a strictly larger sum.

Proof

Write $m = \bar{x}_S$ for the mean.

Step 1: the deviations from the mean add to zero. For each coordinate j, by the definition of the mean, $|S| \, m[j] = \sum_{i \in S} x_i[j]$. So

$$\sum_{i \in S} \big(x_i[j] - m[j]\big) = \sum_{i \in S} x_i[j] - |S| \, m[j] = 0.$$

Step 2: one coordinate. Fix j and write $x_i[j] - \mu[j] = \big(x_i[j] - m[j]\big) + \big(m[j] - \mu[j]\big)$. Squaring,

$$\big(x_i[j] - \mu[j]\big)^2 = \big(x_i[j] - m[j]\big)^2 + 2\big(x_i[j] - m[j]\big)\big(m[j] - \mu[j]\big) + \big(m[j] - \mu[j]\big)^2.$$

Add this over $i \in S$. The factor $m[j] - \mu[j]$ does not depend on i, so the middle terms add to $2\big(m[j] - \mu[j]\big) \sum_{i \in S} \big(x_i[j] - m[j]\big)$, which is 0 by Step 1. The last term is the same for every i, so it adds to $|S|\big(m[j] - \mu[j]\big)^2$. Hence

$$\sum_{i \in S} \big(x_i[j] - \mu[j]\big)^2 = \sum_{i \in S} \big(x_i[j] - m[j]\big)^2 + |S|\big(m[j] - \mu[j]\big)^2.$$

Step 3: all coordinates. Add the equation of Step 2 over $j = 0, \ldots, p-1$. By the definition of the squared distance, the left side becomes $\sum_{i \in S} \|x_i - \mu\|^2$, the first term on the right becomes $\sum_{i \in S} \|x_i - m\|^2$, and the second becomes $|S| \, \|m - \mu\|^2$. This is the equation of the claim.

Step 4: the smallest sum. The term $|S| \, \|m - \mu\|^2$ is a sum of squares times $|S| \geq 1$, so it is at least 0, and it is 0 only when $\mu[j] = m[j]$ for every j, that is, when $\mu = m$. So the sum is smallest at $\mu = m$, and strictly larger at every other μ.

A check. For $S = \{0, 2\}$ and the east coordinates 4 and 2 alone, the mean is 3. With $\mu = 4$, the left side is $0^2 + 2^2 = 4$ and the right side is $(1^2 + 1^2) + 2 \cdot 1^2 = 4$.

Claim 1 is where the mean, the most common summary in statistics, enters an algorithm: among all points, the mean has the smallest sum of squared distances to the data. The claim depends on the *squared* distance. If each point contributed the plain distance instead, the best center would in general not be the mean (for one measurement it is a median), and the learning step would need a different rule.

Two aims become one. Let $\bar{x}$ be the mean of all n points, and let every nonempty cluster have its mean as its center. Claim 1 with $S = C_k$ and $\mu = \bar{x}$, added over the nonempty clusters, gives

$$\sum_{i=0}^{n-1} \|x_i - \bar{x}\|^2 = W + \sum_{k:\ C_k \neq \{\}} |C_k|\, \|\bar{x}_{C_k} - \bar{x}\|^2.$$

The left side does not depend on the split. So making W small, which is cohesion, is the same as making the last sum large, and that sum is large when the cluster means lie far from the overall mean, which is separation. For survey A and split (i), $\bar{x} = (5, 5)$ and $96 = 18 + 3 \cdot 13 + 3 \cdot 13$.

Neither step increases W

Each step finds the best choice for its own part

Claim 2 (no step increases W)

Let z be an assignment and μ centers.

(a) If the partition step, with centers μ, turns z into z', then $W(z', \mu) \le W(z, \mu)$. If some label changed, the inequality is strict.

(b) If the learning step turns μ into μ', then $W(z, \mu') \le W(z, \mu)$.

Proof

Part (a). The centers μ do not change in this step, so we can compare the two sums point by point. Take a point i.

- If $z_i' = z_i$, the point's term $\|x_i - \mu_{z_i}\|^2$ is the same in both sums.
- If $z_i' \neq z_i$, the tie rule says the label changed only because the current center μ_{z_i} is not among the nearest. The new label z_i' is among the nearest, so $\|x_i - \mu_{z_i'}\|^2$ is the smallest of the K squared distances, and it is strictly smaller than $\|x_i - \mu_{z_i}\|^2$.

Adding over all points gives $W(z', \mu) \le W(z, \mu)$. If at least one label changed, at least one term is strictly smaller, so the inequality is strict.

Part (b). The labels do not change in this step, so the clusters $C_0, \ldots, C_{K-1}$ are the same before and after. By W by cluster (Section 3),

$$W(z, \mu) = \sum_{k=0}^{K-1} \sum_{i \in C_k} \|x_i - \mu_k\|^2 \quad \text{and} \quad W(z, \mu') = \sum_{k=0}^{K-1} \sum_{i \in C_k} \|x_i - \mu_k'\|^2.$$

We compare the two sums cluster by cluster.

- If C_k is empty, cluster k contributes nothing to either sum.
- If C_k is nonempty, then $\mu_k' = \bar{x}_{C_k}$, and Claim 1 with $S = C_k$ and the point μ_k gives $\sum_{i \in C_k} \|x_i - \mu_k'\|^2 \le \sum_{i \in C_k} \|x_i - \mu_k\|^2$.

Adding over k gives $W(z, \mu') \le W(z, \mu)$.

In the run on survey A, part (a) is the drop from 36 to 27 when household 1 changed its label, and part (b) is the drop from 112 to 36 and from 27 to 18 when the centers moved to the means.

k-means stops

A quantity that keeps falling and has finitely many values

Claim 3 (k-means stops)

For any points, any K and any starting centers, k-means stops after at most K^n learning steps and at most $K^n + 1$ partition steps.

Proof

Step 1: names. Let $z^{(r)}$ be the labels after partition step r, let $\mu^{(r)}$ be the centers after learning step r, and let $\mu^{(0)}$ be the starting centers. Partition step r uses the centers $\mu^{(r-1)}$. For every round r that has a learning step, let $W_r := W(z^{(r)}, \mu^{(r)})$.

Step 2: W_r falls from round to round. Let $r \geq 1$ and suppose partition step $r + 1$ changes a label, so that learning step $r + 1$ happens. Partition step $r + 1$ turns $z^{(r)}$ into $z^{(r+1)}$ with the centers $\mu^{(r)}$, so Claim 2(a) gives

$$W\big(z^{(r+1)}, \mu^{(r)}\big) < W\big(z^{(r)}, \mu^{(r)}\big) = W_r.$$

Learning step $r + 1$ turns $\mu^{(r)}$ into $\mu^{(r+1)}$ with the labels $z^{(r+1)}$, so Claim 2(b) gives

$$W_{r+1} = W\big(z^{(r+1)}, \mu^{(r+1)}\big) \leq W\big(z^{(r+1)}, \mu^{(r)}\big).$$

Together, $W_{r+1} < W_r$.

Step 3: W_r depends only on the labels. After learning step r, the center of every nonempty cluster of $z^{(r)}$ is the mean of that cluster. By W by cluster,

$$W_r = \sum_{k:\, C_k \neq \{\}} \sum_{i \in C_k} \|x_i - \bar{x}_{C_k}\|^2, \qquad \text{where } C_k = \{i : z_i^{(r)} = k\},$$

because an empty cluster contributes nothing, whatever center it kept. The right side is computed from $z^{(r)}$ alone. So if $z^{(s)} = z^{(t)}$, then $W_s = W_t$.

Step 4: no assignment repeats. Suppose the algorithm has completed R learning steps. For $r = 1, \ldots, R - 1$, partition step $r + 1$ was followed by a learning step, which happens only when that partition step changed a label. So Step 2 applies to every pair of rounds r and $r + 1$ with $1 \leq r < R$, and gives $W_1 > W_2 > \cdots > W_R$. Any two of these numbers differ, so by Step 3 any two of $z^{(1)}, \ldots, z^{(R)}$ differ.

Step 5: counting. There are only K^n assignments, so by Step 4, $R \leq K^n$ whenever R learning steps have been completed. So the algorithm cannot complete $K^n + 1$ learning steps. Partition step $r + 1$ runs only after learning step r, so if R learning steps run, at most $R + 1$ partition steps run, and partition step $R + 1$ stops the algorithm, since otherwise learning step $R + 1$ would follow. So k-means stops after at most K^n learning steps and at most $K^n + 1$ partition steps.

Compare the measure argument of Lecture 2. There, a recursive procedure stops because a nonnegative whole number falls at every call, and such a number cannot fall forever. Here, W_r falls at every round, but it is not a whole number in general (the means can be fractions). What replaces "whole number" is Step 3: W_r is fixed by the assignment, so it can take only as many values as there are assignments.

Common mistake. "W falls at every round, so the algorithm must stop." A falling sequence of numbers can go on forever, as $1, \frac{1}{2}, \frac{1}{4}, \ldots$ does. Steps 3 to 5 are what finish the proof: the finite number of assignments forces the stop.

The bound $K^n + 1$ says little about speed: for 100 points and $K = 2$ it allows more than 10^{30} rounds. On survey A, k-means stopped after three partition steps, and Section 11 returns to the number of rounds.

The approximation gap

How far a stop can be from the best split

Recall that W^* is the value of the k-means problem (Section 3), the smallest $W(\mu)$ over all centers, and that the k-means algorithm returns a W that is at least W^*. An algorithm for the k-means problem has an *approximation guarantee* if there is one fixed number, the same for every input, such that the W it returns is never more than that number times W^*. The ratio of the returned W to W^* is the *approximation gap* on that input. Claim 4 shows that k-means, as stated in Section 4, has no approximation guarantee.

Claim 4 (the approximation gap of k-means is unbounded)

For every number $G > 0$, there are four points and two starting centers for which k-means, with $K = 2$, stops at a split whose W is more than G times W^*.

Proof

Choose a number $L > 1$ with $L^2 > G$. Take four points at $x_0 = (0, 0)$, $x_1 = (0, 1)$, $x_2 = (L, 0)$ and $x_3 = (L, 1)$, a rectangle of width L and height 1, and the starting centers $\mu_0 = x_0$ and $\mu_1 = x_1$.

Step 1: round 1, partition step. Point 0 is at squared distance 0 from μ_0 and 1 from μ_1, so it takes label 0. Point 1 is at 1 and 0, so it takes label 1. Point 2, at $(L, 0)$, is at L^2 from μ_0 and $L^2 + 1$ from μ_1, so it takes label 0. Point 3, at $(L, 1)$, is at $L^2 + 1$ and L^2, so it takes label 1. Every comparison is strict, so the tie rule is never used.

Step 2: round 1, learning step. $C_0 = \{0, 2\}$ and $C_1 = \{1, 3\}$, so the new centers are the centroids $\mu_0 = (L/2, 0)$ and $\mu_1 = (L/2, 1)$.

Step 3: round 2, partition step. Points 0 and 2 are at squared distance $(L/2)^2 = L^2/4$ from μ_0 and $L^2/4 + 1$ from μ_1, so they keep label 0. Points 1 and 3 are at $L^2/4 + 1$ from μ_0 and $L^2/4$ from μ_1, so they keep label 1. No label changes, so k-means stops.

Step 4: W at the stop. Each of the four points is at squared distance $L^2/4$ from its center, so the stop has $W = 4 \cdot L^2/4 = L^2$.

Step 5: an upper bound on W^*. The split $\{0, 1\}$ and $\{2, 3\}$, with centers $(0, \frac{1}{2})$ and $(L, \frac{1}{2})$, puts each point at squared distance $(\frac{1}{2})^2 = \frac{1}{4}$ from its center, so its W is $4 \cdot \frac{1}{4} = 1$. These centers are also each point's nearest, so $W(\mu) = 1$ for them, and since W^* is the smallest $W(\mu)$ over all centers, $W^* \leq 1$.

Step 6: W^* is positive. For any two centers, the four points are all different, and at most two of them can coincide with a center, so at least two points pay a positive amount and $W(\mu) > 0$. The smallest value is attained: W^* is also the smallest $W(z, \mu)$ (Section 3), there are finitely many assignments, and by Claim 1 the best centers for each are the cluster means. So $W^* > 0$.

Step 7: the gap. By Steps 4 and 5, the stop's W divided by W^* is at least $L^2/1 = L^2$, and $L^2 > G$ by the choice of L.

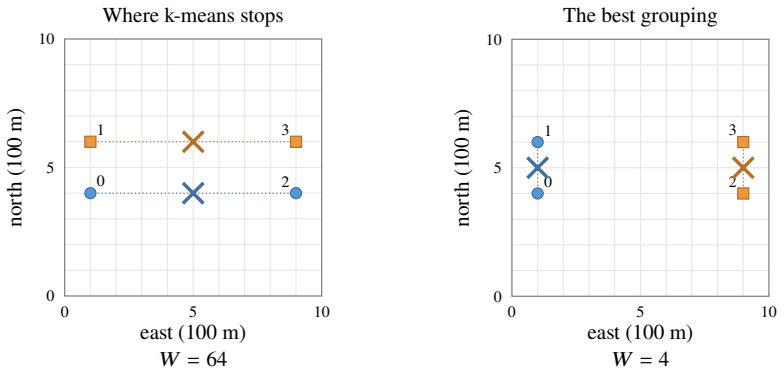


Figure 4. Claim 4 with $L = 4$, drawn twice as large and shifted: $W = 64$ at the stop, against $W^* = 4$.

Figure 4 shows the proof's rectangle with $L = 4$, scaled by 2 and shifted, so that every W is 4 times larger: k-means stops at $W = 64$, while $W^* = 4$, a gap of $16 = L^2$. Stretching the rectangle makes the gap as large as we like.

What a stop does guarantee. When k-means stops, the last partition step changed nothing, so every point's center is one of its nearest centers. The last learning step moved every nonempty cluster's center to its mean, and the labels have not changed since. So running either step again changes nothing. This does not mean that no better split is close by. In Figure 4, moving point 1 alone to cluster 0 and recomputing the means lowers W from 64 to $136/3$, about 45.3. The partition step never makes this move, because with the centers $(5, 4)$ and $(5, 6)$ point 1 is already at a nearest center.

What restores a guarantee. The gap depends on the starting centers: from x_0 and x_2 instead, k-means reaches W^*. Software therefore runs k-means from several starting centers and keeps the run with the smallest W (in scikit-learn, the argument `n_init` of `KMeans` sets the number of runs), which helps in practice but gives no guarantee. The k-means++ rule for choosing starting centers does give one. It picks the starting centers at random among the points, favoring points far from the centers already picked (Section 11), and before any round the expected W is at most $8(\ln K + 2)$ times W^* (Arthur and Vassilvitskii, 2007). By Claim 2, the rounds of k-means never increase W, so k-means started from k-means++ centers keeps this guarantee in expectation.

7 Two surveys, one algorithm

The same algorithm on different points

In this section, we move from the six households to the question that motivated them. Up to now, the six households of survey A were the whole input, and the question was whether k-means returns a good split of those six. The question in Section 1 was broader: where to park the vans for all the households of the city. A department rarely knows where every household is, and often plans from a survey of some of them.

To see what this changes, suppose the city has $N = 24$ households, and that two surveys, A and B, each record six different households, drawn at random from the city. Figure 5 shows the three data sets. We run k-means with $K = 2$ on each, starting from the points of the first two households in each list.

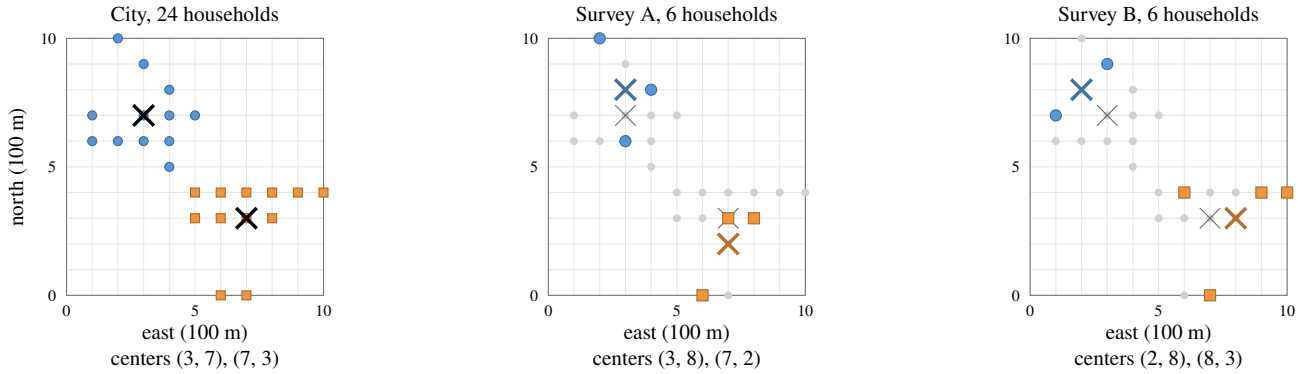


Figure 5. Black crosses: the city's centers, repeated as thin crosses in the survey panels. Colored crosses: each survey's centers.

Data	Households	Centers from k-means	W	Best split of these households?
Survey A	6	$(3, 8)$ and $(7, 2)$	18	yes, of 31 splits
Survey B	6	$(2, 8)$ and $(8, 3)$	26	yes, of 31 splits
City	24	$(3, 7)$ and $(7, 3)$	90	yes, of 8,388,607 splits

In all three runs, k-means stops at the best split of its own input, which we checked by computing W for every split into two nonempty clusters (for n households there are $2^{n-1} - 1$ of them: 31 for six households and 8,388,607 for 24). In each case the best split is the only one with the smallest W. So each run returns the exact answer to the k-means problem for the data it receives. The three answers still differ, even read as pairs in either order. The difference does not come from the algorithm. It comes from the households: survey B, for example, saw two households from the upper-left group and four from the lower-right group.

Definition 4 (population, sample, and three kinds of centers)

The *population* is the set of all households the question is about, here the city. A *sample* is the set of households whose data we have, here one survey's six. The *population centers* are the centers of the best split of the whole population, here $(3, 7)$ and $(7, 3)$. The *sample centers* are the centers of the best split of the sample. The *reported centers* are the centers that k-means returns for the sample. Centers are compared as a pair, in either order, because the names of the labels carry no meaning.

In words, the population centers are where the vans should park for the whole city, and the reported centers are what the department can compute from a survey.

Two gaps between what we compute and what we want

One from the algorithm, one from the sample

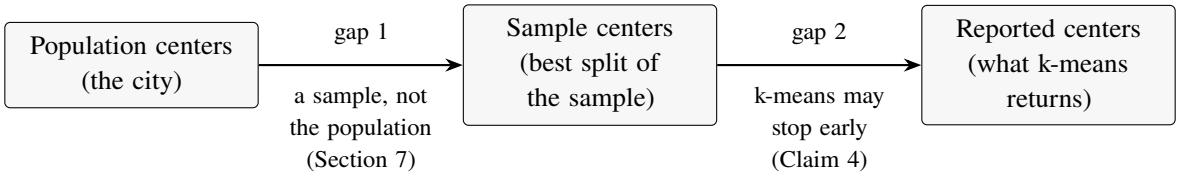


Figure 6. Two reasons why the centers we report can differ from the population centers.

Figure 6 separates two reasons why reported centers can differ from the population centers. Gap 2 is about the algorithm: k-means can stop at a split that is not the best for the sample, as in Claim 4. Running k-means from several starting centers narrows this gap, and Claims 2 to 4 describe it. Gap 1 is about the data: the sample centers need not equal the population centers. Both surveys close gap 2 completely, and gap 1 remains. No change to the algorithm can close gap 1, because the algorithm receives only the sample.

The same gap in real data. Seymour et al. (2019) applied a version of k-means to 29 clinical variables from 20,189 hospital encounters for sepsis and reported four groups of patients, with 28-day mortality from 5% to 40%. Yoon et al. (2026) returned to these groups with 48,246 emergency department encounters from Stockholm, Oxford and Oslo. They gave each patient the label of the nearest of the four reported centers, which is the partition step with given centers, and they also ran consensus k-means with $K = 4$ at each site. The two labelings agreed little: a measure of agreement called Cohen's kappa, which is 1 for perfect agreement and 0 for the agreement expected by chance, was between 0.32 and 0.40, and the phenotypes found at the three sites also disagreed with one another. The authors conclude that the four phenotypes are not generalizable across these cohorts. Part of such a disagreement can come from sampling, as with our two surveys, and part from real differences between the patients of different hospitals and countries. Telling these apart is again a question of statistics.

The simplest case. With $K = 1$, every household is in the same cluster, and Claim 1 says that the best center is the mean. So the population center is the mean of the 24 city households, $(5, 5)$, and the sample center is the mean of a survey's six households. Asking how close a sample mean is to the population mean, and how much it changes from one sample to the next, are the first questions of the next part of the course.

A final caution concerns interpretation. A cluster is a set of households that are close on the map. Whether parking a van at its center actually improves vaccination or screening is a question about effects, not about distances, and we return to such questions in the causal part of the course.

8 Snow's map

Centers given, and centers computed

In this section, we read a well-known public-health map with the tools of this lecture, and we mark where those tools stop. In late August and early September 1854, cholera killed more than 600 people in Soho, London: John Snow's table lists 616 deaths between 19 August and 30 September. Snow, a physician who held that cholera spread through drinking water, suspected the public pump in Broad Street. He put his case to the parish Board of Guardians on 7 September, and the handle of the pump was removed on 8 September. By then the daily number of new fatal cases had already fallen, from 143 on 1 September to 12 on 8 September, and Snow wrote that the attacks "had so far diminished before the use of the water was stopped, that it is impossible to decide whether the well still contained the cholera poison" (Snow, 1855a). His map of the deaths came afterwards. It was first shown in December 1854 and printed in 1855, so it presented and tested a hypothesis that Snow already held, rather than leading him to it (Brody et al., 2000).

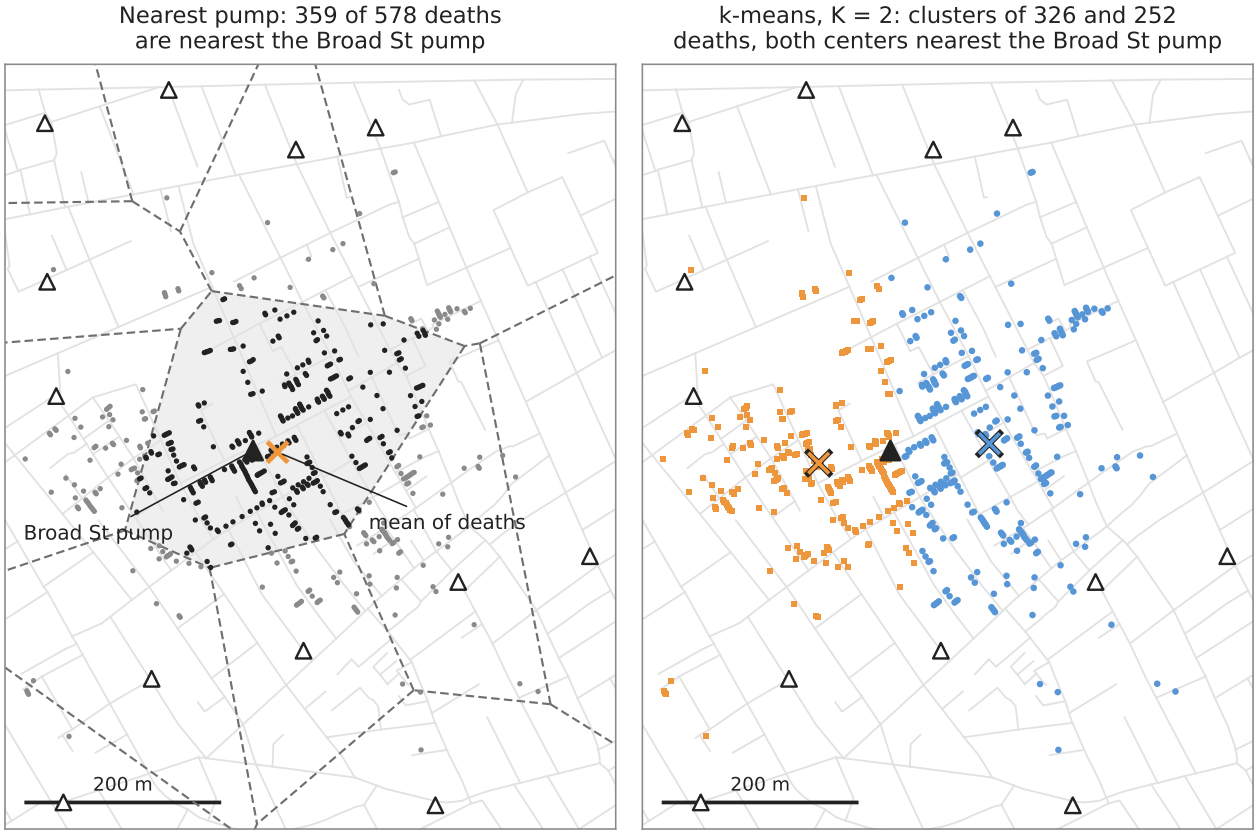


Figure 7. Snow's map, digitized. Left: nearest pump by straight-line distance, with the Broad Street region shaded. Right: k-means with $K = 2$ on the deaths alone.

Figure 7 uses a digitized version of the map, with 578 deaths and the 13 public pumps.[1]

[1]Data: `Snow.deaths`, `Snow.pumps` and `Snow.streets` in the R package HistData, digitized in 1992 by R. Dodson and W. Tobler from the map reprinted in *Snow on Cholera* (1936). Several deaths at one house are drawn as a short stack, as on Snow's map, so a point marks a stack position rather than an exact door. One map unit is about 55 m, which we estimated by matching the pumps to a georeferenced

Centers given: the partition step. We take the 13 pumps as fixed centers and give each death the label of its nearest pump. This is the partition step of k-means without the learning step, because the pumps cannot move. The regions of points nearest each pump form a *Voronoi diagram* of the pumps (the dashed lines in Figure 7). By straight-line distance, 359 of the 578 deaths are nearest the Broad Street pump. Snow drew the edge of this region himself, in a second version of the map for a parish inquiry in 1855:

> "The inner dotted line on the map shews the various points which have been found by careful measurement to be at an equal distance by the nearest road from the pump in Broad Street and the surrounding pumps [. . .] the deaths either very much diminish, or cease altogether, at every point where it becomes decidedly nearer to send to another pump than to the one in Broad Street. At these points I ascertained that the people did generally send to the pump which was nearer." (Snow, 1855b, p. 109)

The passage carries two lessons for this lecture. First, the distance is a modeling choice. People walked to a pump along streets, so Snow measured along streets, and he drew one region, not a full Voronoi diagram (Koch, 2004). The *walking distance* between two places is the length of a shortest route through the street network, a graph problem of the kind in Lectures 5 and 6. In our approximate computation on the digitized streets, 91 of the 578 deaths (about one in six) have a different nearest pump by walking distance than by straight-line distance. Second, the label from the partition step is a model of where each household drew its water, and Snow tested the model by asking. He also reported where it failed: the water of the pump in Marlborough Street was so impure that many people living near it fetched water from Broad Street instead.

Centers computed

What the mean of the deaths shows, and what it does not

One center. We now set the pumps aside and run k-means on the deaths alone. With $K = 1$, Claim 1 says that the center is the mean of the deaths. It lies about 25 m from the Broad Street pump, while the next nearest pump is about 210 m away. This looks like strong evidence, but where a mean falls depends on where the map's edges are. Points in an *even spread* over the smallest rectangle that contains the deaths have their mean about 30 m from the pump, because the map is drawn around the outbreak. Even the 219 deaths that are nearer another pump have their mean about 35 m from the Broad Street pump, because they lie on all sides of it. The mean shows where the deaths balance, and here that is near the middle of the map.

What separates the deaths from an even spread is how tightly they gather around the pump: 225 of the 578 deaths (39%) lie within 100 m of it, against about 10% of an even spread over the same rectangle. Even this comparison is incomplete. Without a source, we would expect deaths to be spread evenly only if people lived evenly over the map, and the map does not show how many people lived in each house. Snow knew this: he explained the smaller number of deaths in the south half of Poland Street by its being "less densely inhabited" (Snow, 1855b, p. 110). A study that estimated the number of residents of every house found the highest death rates next to the Broad Street pump, and found that the workhouse in Poland Street, which shows a group of deaths on the map, had a low death rate once its several hundred residents were counted (Shiode et al., 2015). The number of people at risk is the *denominator* of a rate, and deciding what an observed pattern should be compared with is a question of statistics.

Two centers. With $K = 2$, k-means cuts the outbreak into two clusters, of 326 and 252 deaths, and both centers are nearer the Broad Street pump than any other pump (Figure 7, right). Two clusters do not mean two sources.

copy of the map (the package documentation gives 100 m, which is not correct). The figure and every number in this section are produced by `make-lec10-snow-figure.py` and checked by `verify-lec10.py`.

k-means returns K clusters for any data, including data with no groups at all, so a center summarizes where points lie and does not locate a cause.

What would show a cause. Whether the water of the Broad Street pump caused the deaths is a third kind of question, which no summary of this map settles. Snow's strongest evidence came from South London, where the pipes of two water companies ran down the same streets ("the mixing of the supply is of the most intimate kind"). One company drew its water from the Thames where the river carried London's sewage, and the other had moved its intake upstream, so that the households were, in Snow's words, "divided into two groups without their choice". In the first seven weeks of the 1854 epidemic, houses supplied by the first company had 315 cholera deaths per 10,000 houses, against 37 for the second (Snow, 1855a). We return to this comparison, a natural experiment, in the causal part of the course. Retellings in which the map revealed the pump, or in which removing the handle ended the outbreak, do not match Snow's own account (Brody et al., 2000).

From algorithms to statistics. In the algorithms part of the course, the input was fixed, and an algorithm was correct when it returned the defined answer for that input. From here on, the input is often a sample from a larger population, and the answer we want is defined on the population, or concerns a cause. A correct algorithm is then not enough. We also need to say how far its answer can be from the population answer and how much it changes from one sample to another (Section 7), and what a pattern should be compared with before it counts as evidence (this section).

9 Summary

What we computed, and what we proved

In this section, we collect the answers of the lecture in one table.

Question	Answer
What is clustered?	n points $x_0, \ldots, x_{n-1}$, each with p measurements (in the example, households on a map).
What is a clustering?	Nonempty groups of points, no two overlapping (Definition 1). Ideally it is well separated, but one big cluster and n single ones always are, so a method must choose a criterion Q, and methods such as DBSCAN choose differently.
What does k-means assume?	A fixed K, every point in one cluster, one center per cluster (Definition 2), and the squared distance, so that a split is scored by W (Definition 3).
Why not try every split?	There are K^n assignments, which grows exponentially in n.
What problem does k-means aim at?	The k-means problem: where to put K centers so that the total squared distance from each point to its nearest center, $W(\mu)$, is smallest. Its value is W^* (Definition 3), and finding it exactly is hard.
What does the k-means algorithm do?	It returns an approximation to that answer. It alternates two parts: partition the points by their nearest centers, then learn new centers as the cluster means (Section 4).
Why the mean?	For one cluster, the mean gives the smallest sum of squared distances (Claim 1).
Does W go down?	Neither step increases it, and a label change lowers it strictly (Claim 2).
Does k-means stop?	Yes, after at most $K^n + 1$ partition steps, because W falls at every round and takes finitely many values (Claim 3).
Is the stop the best split?	Not always, and the gap is unbounded: for every G there are inputs where the stop has more than G times W^* (Claim 4). k-means++ starting centers give a guarantee in expectation.
Is the best split of a sample the population answer?	Not in general. Different samples give different centers, even when k-means returns the best split of each (Section 7).
Does a center locate a cause?	No. On Snow's map, the mean of the deaths lies near the Broad Street pump partly because the map is drawn around it, and $K = 2$ cuts one outbreak in two (Section 8).

The first ten rows are about a question, an algorithm and its guarantees. The last two rows are about data: how an answer computed from a sample relates to the population the sample comes from, and what a pattern must be compared with before it counts as evidence. The rest of the course takes up these questions.

10 Exercises

Runs, ties, starts, units, samples and Snow's map

The exercises practice a run by hand, the role of the tie rule and the starting centers, the units of the measurements, and Sections 7 and 8.

1. Six households lie along one street, at east coordinates 0, 2, 5, 6, 8 and 9, so $p = 1$. Run k-means with $K = 2$ and starting centers 0 and 2. Give the labels and centers after each round, and the final W.

2. Claim 3 needs the tie rule. Explain why, by describing what could go wrong if a point whose current center is among the nearest could switch to another nearest center.

3. For the four points of Figure 4, there are six ways to choose two of them as starting centers. For each choice, find where k-means stops and its W.

4. In the example of Figure 4, suppose the second coordinate is recorded in units of 10 m instead of 100 m, so every second coordinate is multiplied by 10, while the first stays in units of 100 m. Which of the two splits in Figure 4 has the smaller W now? What does this say about the units of the measurements?

5. What is the smallest possible W when $K = n$? Explain why choosing K to make the smallest W as small as possible is not a useful rule.

6. With $K = 1$, compute the center for survey A, for survey B and for the city (the city's two clusters each have 12 households, with means $(3, 7)$ and $(7, 3)$). Survey A's center equals the city's. Does it follow that survey A's two centers with $K = 2$ equal the city's two centers?

7. Snow measured the distance to a pump along the streets. (a) Describe the street network as a graph, saying what the vertices and edges are, and say which search from Lectures 5 and 6 finds the route with the fewest street segments. Why is that route not always the shortest one? (b) With walking distance in place of the squared straight-line distance, is the mean of a cluster still its best center? Which step of the proof of Claim 1 no longer applies?

8. In the proof of Claim 4, take $L = 10$. At what W does k-means stop from the starting centers x_0 and x_1, and what is W^*? Which pairs of points, used as starting centers, lead k-means to W^*?

11 Beyond the lecture

Cost, hardness, populations, no perfect criterion, DBSCAN and mixtures

In this section, we sketch where the lecture's topics lead.

The cost of one round. The partition step computes nK squared distances, and each takes p subtractions, p squarings and $p - 1$ additions. The learning step adds up each coordinate of each point once and divides Kp sums. So the number of basic steps in one round lies between two positive constant multiples of nKp, that is, it is $\Theta(nKp)$ with all three sizes large. The number of rounds is a different matter. Claim 3 allows up to $K^n + 1$, and there are sets of points in the plane, with K growing with n, on which k-means from some starting centers needs a number of rounds that grows exponentially in n (Vattani, 2011). In practice few rounds are usually needed. The k-means problem itself (Definition 3) is computationally hard. It is NP-hard, already for $K = 2$ when p can grow (Aloise et al., 2009): an algorithm that always returned the best split in a number of steps bounded by a polynomial in n and p would also give such an algorithm for a large family of problems for which none is known.

Choosing starting centers. The k-means++ rule picks the first starting center uniformly at random among the points, and each further one at random with probability proportional to the squared distance from a point to the nearest center already chosen. Section 6 states the guarantee that this rule gives, where the expected W means the average of W over the random choices.

Populations as distributions. The city of Section 7 stands for a population that is usually far larger than any sample, and statistics models it as a probability distribution from which points are drawn at random. The population centers are then the centers that make the *expected* squared distance from a random point to the nearest center (its average over the distribution) as small as possible. Pollard (1981) showed that, under two conditions (the squared length of a random point has a finite expected value, and for every number of centers up to K the best centers are unique), the sample centers of n random points approach the population centers with probability one as n grows. This is a first example of a statistical guarantee: a statement about how an answer computed from a sample behaves as the sample grows.

No perfect criterion. Kleinberg (2002) stated three properties that seem natural for any rule that turns dissimilarities into a clustering, and proved that no rule has all three.

(i) Multiplying every dissimilarity by the same positive number does not change the clustering.

(ii) Every clustering that places all points can be produced by some dissimilarity.

(iii) Shrinking within-pair dissimilarities while enlarging between-pair ones does not change the clustering.

So the choice of criterion in Definition 1(d) is a real choice, not a gap that a better method will close.

Density instead of centers: DBSCAN. DBSCAN (Ester et al., 1996) takes two numbers in place of K: a radius and a minimum count. A point is a *core* point if at least the minimum count of points, itself included, lie within the radius of it. Join two core points by an edge when they lie within the radius of each other. The clusters are then the connected components of this graph, which a graph search such as BFS (Lectures 5 and 6) finds, and a point that is not a core point but lies within the radius of one joins the cluster of such a core point. Every other point is noise. So the number of clusters is an output, a cluster can have any shape, and some points belong to no cluster. The result depends on the radius and the minimum count, much as the result of k-means depends on K.

Mixtures. A Gaussian mixture model describes each group of points by its own normal distribution and assigns each point a probability of belonging to each group, rather than one label. Its usual fitting algorithm, expectation

maximization, alternates two steps much like k-means, and k-means is its limit when every group is a round normal distribution with a common variance that shrinks to zero (Bishop, 2006, Section 9.3.2).

Further reading. James et al. (2023, Section 12.4.1, printed pages 515 to 519) define the within-cluster variation and give the algorithm, with figures of several runs from different starting assignments. Their within-cluster variation adds squared distances over all pairs of points in a cluster, and their identity (12.18) shows that it equals twice the cluster's contribution to W at its mean. Their algorithm starts from random labels and so runs the learning step first. Rudin (n.d.) presents k-means as alternating minimization of one objective, as in Claim 2. On Snow's map, Brody et al. (2000) separate the history of the Broad Street outbreak from its legend, Koch (2004) discusses the claim that Snow drew the first Voronoi diagram, and Shiode et al. (2015) estimate death rates house by house. The R package `cholera` by Peter Li (`https://github.com/lindbrook/cholera`) reanalyzes the map with walking distances.

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